ASSIGNMENT
Data Quality and Validation in ETL

Q1 Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?
ANS- Data Quality refers to how fit the data is for its intended use.
High-quality data is:
● Accurate
● Complete
● Consistent
● Unique
● Valid
● Timely
In ETL:
Data quality checks ensure bad data does not enter the target system.

It's more than just data cleaning because:

- Data cleaning only fixes surface-level issues like duplicates or missing values.
- Data quality covers end-to-end validation — checking source integrity, transformation logic, schema consistency, and business rule compliance.
- It involves continuous monitoring to detect errors, maintain timeliness, and ensure that the data supports meaningful insights.

Q 2 : Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Ans-

Issue	Explanation	Effect on Dashboards & Decisions
Inaccurate Data	Errors in values (wrong numbers, typos, miscalculations).	KPIs show false trends → managers act on wrong signals.
Incomplete Data	Missing fields or records.	Dashboards under-represent reality → decisions ignore key factors
Inconsistent Data	Different formats or conflicting entries across sources.	Charts misalign → comparisons become unreliable.
Outdated Data	Delays in ETL or stale sources.	Dashboards reflect past conditions → decisions lag behind market.
Invalid Data	Data not following business rules (e.g., negative age).	Dashboards highlight impossible scenarios → erodes trust.

Q3 : What is duplicate data? Explain three causes in ETL pipelines
ANS- Duplicate data refers to records that appear more than once in a dataset, either fully identical or partially overlapping. In ETL pipelines, duplicates distort analytics, inflate KPIs, and waste storage/resources.

Cause	Explanation	Example in ETL
Multiple Source Systems	Same customer or transaction captured in different applications	CRM and e-commerce both record the same purchase.
Faulty Extraction Logic	Improper joins or repeated pulls during extraction.	ETL job re-runs without deduplication, pulling the same rows twice.
Human / Input Errors	Manual entry mistakes or inconsistent identifiers.	User registers twice with slightly different email IDs.

Q4 : Differentiate between exact, partial, and fuzzy duplicates.
ANS- Duplicate data means having the same or similar records more than once in a dataset.
There are three types:

- **Exact duplicates**: Records that are completely the same in all columns. Example – two rows with the same customer ID, name, and purchase details.
- **Partial duplicates**: Records that match in some columns but differ in others. Example – same customer ID and name but different purchase amount.
- **Fuzzy duplicates**: Records that look similar but are not exactly the same, often because of spelling or formatting differences. Example – "Rahul Mehta" and "R. Mehta" or "Mumbai" and "Bombay."

Q 5 : Why should data validation be performed during transformation rather than after loading
Ans- Data validation means checking if the data is correct, complete, and follows rules.
It should be done during transformation because:

- Errors can be caught early before they enter the warehouse.
- Prevents bad data from polluting dashboards and reports.
- Easier and cheaper to fix problems at transformation than after loading.
- Ensures only clean and reliable data is loaded, so decisions are based on truth.

Q6 : Explain how business rules help in validating data accuracy. Give an example

ANS-. Business rules are conditions set by the organization to ensure that data makes sense in the real business context. They help check whether the data is not only technically correct but also logically valid for decision-making.

- **How they help:** Business rules act like filters. They catch values that may be valid in format but wrong in meaning.
- **Example:** A business rule may say *Customer age must be between 0 and 120.*
    - If the ETL pipeline receives an age of 250, the rule will flag it as invalid even though the number format is correct.

Q 7 : Write an SQL query on to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount)

```sql
use  Sales_transaction;


CREATE TABLE Transactions (
    Txn_ID INT,
    Customer_ID VARCHAR(10),
    Customer_Name VARCHAR(50),
    Product_ID VARCHAR(10),
    Quantity INT,
    Txn_Amount INT,
    Txn_Date DATE,
    City VARCHAR(50)
```

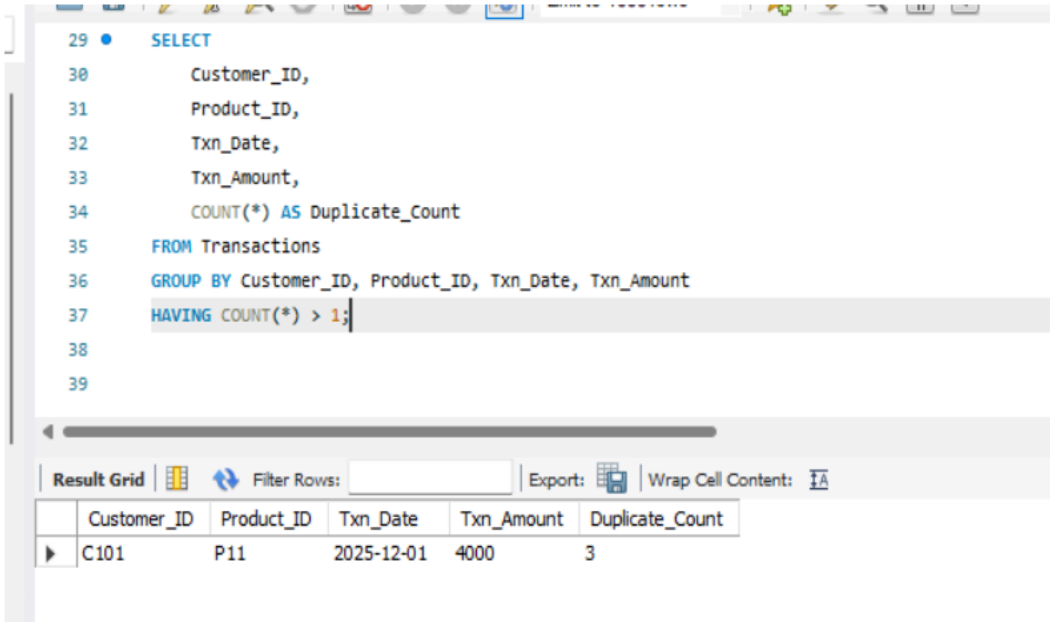
Limit to 1000 rows

```sql
INSERT INTO Transactions (Txn_ID, Customer_ID, Customer_Name, Product_ID, Quantity, Txn_Amount, Txn_Date, City) VALUES
(201, 'C101', 'Rahul Mehta', 'P11', 2, 4000, '2025-12-01', 'Mumbai'),
(202, 'C102', 'Anjali Rao', 'P12', 1, 1500, '2025-12-01', 'Bengaluru'),
(203, 'C101', 'Rahul Mehta', 'P11', 2, 4000, '2025-12-01', 'Mumbai'),
(204, 'C103', 'Suresh Iyer', 'P13', 3, 6000, '2025-12-02', 'Chennai'),
(205, 'C104', 'Neha Singh', 'P14', NULL, 2500, '2025-12-02', 'Delhi'),
(206, 'C105', 'N/A', 'P15', 1, NULL, '2025-12-03', 'Pune'),
(207, 'C106', 'Amit Verma', 'P16', 1, 1800, NULL, 'Pune'),
(208, 'C101', 'Rahul Mehta', 'P11', 2, 4000, '2025-12-01', 'Mumbai');

select* from Transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Txn_ID	Customer_ID	Customer_Name	Product_ID	Quantity	Txn_Amount	Txn_Date	City
201	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
202	C102	Anjali Rao	P12	1	1500	2025-12-01	Bengaluru
203	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
204	C103	Suresh Iyer	P13	3	6000	2025-12-02	Chennai
205	C104	Neha Singh	P14	NULL	2500	2025-12-02	Delhi
206	C105	N/A	P15	1	NULL	2025-12-03	Pune
207	C106	Amit Verma	P16	1	1800	NULL	Pune
208	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai



```sql
SELECT
    Customer_ID,
    Product_ID,
    Txn_Date,
    Txn_Amount,
    COUNT(*) AS Duplicate_Count
FROM Transactions
GROUP BY Customer_ID, Product_ID, Txn_Date, Txn_Amount
HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Product_ID	Txn_Date	Txn_Amount	Duplicate_Count
C101	P11	2025-12-01	4000	3

Question 8 : Enforcing Referential Integrity Assume the following CUSTOMERS_MASTER table:

Identify Sales_Transactions.Customer_ID values that violate referential integrity when joined with Customers_Master and write a query to detect such violations

ANS- SELECT s.Customer_ID

FROM Sales_Transactions s

LEFT JOIN Customers_Master c

ON s.Customer_ID = c.Customer_ID

WHERE c.Customer_ID IS NULL;

Output

Violating_Customer_ID

C105

C106